

Classical Demand II: Three Bridges — Shephard, Slutsky, and Roy

Advanced Microeconomics 1: Economic Decision Making. Lecture 3

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Where we left the question

- **Lecture 2 built the ideal answer:** the correct indexation payment for household h is $e^h(p^1, u^0) - w$.
- **And admitted the catch:** e , h , v are all constructed from u — and no dataset contains u . The answer is ideal and impossible :(.
- **What the data does contain:** Marshallian demand $x(p, w)$ — purchases at prices and budgets. Supermarket scanners, the Household Budget Survey, etc.

Today: bridging from the observables x , to the unobservables we actually want.

Bridge 1: Shephard

Shephard intuition

How much does your expenditure change if the price of good k changes by 1 cent?



Ronald Shephard, 1912-1982

Shephard's lemma

Maintained assumptions: u continuous and LNS, $\succsim$ is strictly convex, and h is single-valued and differentiable.

Theorem · Shephard's lemma (MWG 3.G.1)

For all $p \gg 0$ and u :
$$h(p, u) = \nabla_p e(p, u).$$

Proof → p. 22

- Raise p_ℓ by a cent, and the cost of staying as well off rises by — how much you buy of good ℓ .
- The true cost-of-living index's slope is (Hicksian) demand.

Properties of h we now know

Theorem · MWG 3.G.2

With $h(\cdot, u)$ continuously differentiable at (p, u) , the $L \times L$ matrix $D_p h(p, u)$ satisfies:

- (i) $D_p h(p, u) = D_p^2 e(p, u)$;
- (ii) $D_p h(p, u)$ is **negative semidefinite** and **symmetric**;
- (iii) $D_p h(p, u) p = 0$.

Proof → p. 23

- (ii) NSD is the differential compensated law of demand — it is exactly **the concavity of e** .
- (ii) **symmetry** is new information: cross-effects of compensated demand are equal, $\partial h_\ell / \partial p_k = \partial h_k / \partial p_\ell$.

Shepard buys us: CV and EV are areas under Hicksian demand

One price changes: $p_1^0 \rightarrow p_1^1$, others fixed at p_{-1} . Since $h_1 = \partial e / \partial p_1$ (Shephard), the fundamental theorem of calculus gives

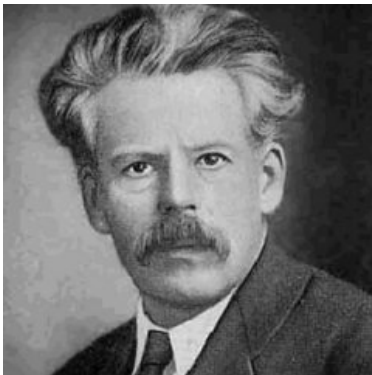
$$-CV = e(p^1, u^0) - e(p^0, u^0) = \int_{p_1^0}^{p_1^1} h_1(t, p_{-1}, u^0) dt, \quad -EV = \int_{p_1^0}^{p_1^1} h_1(t, p_{-1}, u^1) dt.$$

- The answer is **an area under a compensated demand curve**.

Bridge 2: Slutsky

Slutsky intuition

How does (Hicksian) demand for l change with prices of good p ?



Eugen Slutsky, 1880-1948

Slutsky derivation

Fix (p, w) and let $u = v(p, w)$; duality gives that p :

$$h_\ell(p, u) = x_\ell(p, e(p, u)).$$

Differentiate both sides with respect to p_k (chain rule on the second argument):

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, e(p, u))}{\partial p_k} + \frac{\partial x_\ell(p, e(p, u))}{\partial w} \cdot \frac{\partial e(p, u)}{\partial p_k}.$$

Then as by Shephard, $\partial e / \partial p_k = h_k(p, u)$; and at $u = v(p, w)$ we have $e(p, u) = w$ and $h_k(p, u) = x_k(p, w)$. Hence

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w) = s_{\ell k}(p, w).$$

The Slutsky equation

Theorem · Slutsky equation (MWG 3.G.3)

For all (p, w) and $u = v(p, w)$:

$$\underbrace{\frac{\partial h_\ell(p, u)}{\partial p_k}}_{\text{unobservable}} = \underbrace{\frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w)}_{\text{estimable from demand data}}$$

Proof → p. 24

- Denote $D_p h(p, u) = S$.
- h is unobservable, yet its derivative matrix is computable from Marshallian data which is observable!
- The substitution effect for proportional prices changes is 0: $D_p h \times p = 0$.

Slutsky decomposition

$$\frac{\partial x_\ell}{\partial p_k} = \underbrace{\frac{\partial h_\ell}{\partial p_k}}_{\text{Substitution}} - \underbrace{\frac{\partial x_\ell}{\partial w}}_{\text{Income}} x_k$$

- Demand response to a price change = substitution effect + income effect.

Substitutes and complements

Definition · net substitutes / complements

Goods ℓ and k are **(net) substitutes** if $\partial h_\ell / \partial p_k \geq 0$ and **(net) complements** if < 0 .

- Defined on *Hicksian* demand: symmetry makes the relation mutual (ℓ substitutes for $k \iff k$ for ℓ), which the Marshallian “gross” notion fails — asymmetric income effects can make ℓ a gross substitute for k but not conversely.
- From $D_p h p = 0$ and $\partial h_\ell / \partial p_\ell \leq 0$: **every good has at least one net substitute**.
Substitution is not optional in this theory $\Rightarrow$ fixed baskets can never be exactly right.

When the simple area is legal

Practitioners often report *Marshallian* consumer surplus: $\Delta CS = \int_{p_1^1}^{p_1^0} x_1(t, p_{-1}, w) dt$, no h anywhere, easy!

Theorem · no wealth effects (Marshall's case)

If $\partial x_1(p, w)/\partial w = 0$ for all (p, w) in the relevant range, then

$$CV = \Delta CS = EV.$$

Proof → p. 26

- Plausible when the good is a small budget share.

Bridge 3: Roy

Roy's identity intuition

Your demand for good k is how much a price rise in k hurts.



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René Roy, 1984-1977

Roy's identity

Theorem · Roy's identity (MWG 3.G.4)

If v is differentiable at $(p, w) \gg 0$, then for every ℓ :

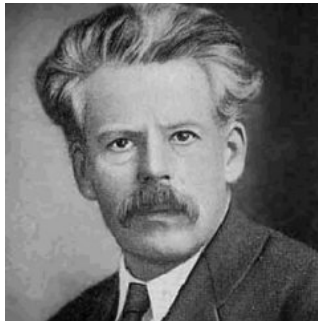
$$x_{\ell}(p, w) = - \frac{\partial v(p, w) / \partial p_{\ell}}{\partial v(p, w) / \partial w}.$$

Proof → p. 25

- $-\frac{\delta v}{\delta p}$ gives the change in value of a change in p_I , intuitively this is closely related to how much of I you buy, x_I .
- But $-\frac{\delta v}{\delta p}$ is measured in utils and x_I in euros. To convert, we need to know how utils vary with w , i.e. $\frac{\delta v}{\delta w}$.
- Roy's identity is a differential equation linking an *estimated* $x(p, w)$ to the unknown v . Solve it, and the data hand you indirect utility — hence e .

What was this all for

Putting things together



- In L2, we derived the cost of inflation by assuming a utility function.
- Now we want to leverage Shephard, Slutsky and Roy to find CV from data, x , only.
- That is exactly what we will do, next lecture!

Where the question stands

Takeaways

- **Three bridges built:** Shephard, Slutsky, and Roy.
- **The answer is estimable:** We can now link the theoretical objects we want (CV, EV) to observable data x .
- **Still to do:** (a) the actual connection and (b) all of this assumed a rational household *exists* behind the data. When does observed demand actually pin one down?

Appendix: proofs

Proof: Shephard's lemma (MWG 3.G.1)

Assume $h(p, u) \gg 0$ single-valued and differentiable. Write $e(p, u) = p \cdot h(p, u)$ and differentiate in p :

$$\nabla_p e(p, u) = h(p, u) + [D_p h(p, u)]^T p.$$

The second term vanishes:

- EMP first-order conditions: $p = \lambda \nabla u(h(p, u))$ for some $\lambda \geq 0$.
- “No excess utility” holds identically in p : $u(h(p, u)) = u$. Differentiate in p :
 $[D_p h]^T \nabla u(h) = 0$.
- Hence $[D_p h]^T p = \lambda [D_p h]^T \nabla u(h) = 0$, and $\nabla_p e = h$. ■

Proof: properties of $D_p h$ (MWG 3.G.2)

- (i) Shephard: $h(p, u) = \nabla_p e(p, u)$ identically. Differentiate in p : $D_p h = D_p^2 e$.
- (ii) $e(\cdot, u)$ is twice continuously differentiable here, so its Hessian is **symmetric** (Young's theorem); and e is **concave** in p (Lecture 2), so its Hessian is **negative semidefinite**. Both properties transfer to $D_p h$ by (i).
- (iii) $h(\cdot, u)$ is homogeneous of degree zero in p (Lecture 2). Differentiate $h(\alpha p, u) = h(p, u)$ in α at $\alpha = 1$ (Euler): $D_p h(p, u) p = 0$. ■

Proof: the Slutsky equation (MWG 3.G.3)

Fix (p, w) and let $u = v(p, w)$; duality gives the identity, in p :

$$h_\ell(p, u) = x_\ell(p, e(p, u)).$$

Differentiate both sides with respect to p_k (chain rule on the second argument):

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, e(p, u))}{\partial p_k} + \frac{\partial x_\ell(p, e(p, u))}{\partial w} \cdot \frac{\partial e(p, u)}{\partial p_k}.$$

Two substitutions finish it: by Shephard, $\partial e / \partial p_k = h_k(p, u)$; and at $u = v(p, w)$ we have $e(p, u) = w$ and $h_k(p, u) = x_k(p, w)$. Hence

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w) = s_{\ell k}(p, w). \blacksquare$$

Proof: Roy's identity (MWG 3.G.4)

Let $\bar{u} = v(\bar{p}, \bar{w})$. Duality gives the identity, in p :

$$v(p, e(p, \bar{u})) = \bar{u}.$$

Differentiate with respect to p_ℓ at $p = \bar{p}$ (chain rule):

$$\frac{\partial v(\bar{p}, \bar{w})}{\partial p_\ell} + \frac{\partial v(\bar{p}, \bar{w})}{\partial w} \cdot \frac{\partial e(\bar{p}, \bar{u})}{\partial p_\ell} = 0.$$

By Shephard, $\partial e / \partial p_\ell = h_\ell(\bar{p}, \bar{u}) = x_\ell(\bar{p}, \bar{w})$ (duality at $\bar{u} = v(\bar{p}, \bar{w})$). Since $\partial v / \partial w > 0$:

$$x_\ell(\bar{p}, \bar{w}) = - \frac{\partial v(\bar{p}, \bar{w}) / \partial p_\ell}{\partial v(\bar{p}, \bar{w}) / \partial w}. \blacksquare$$

Proof: no wealth effects $\Rightarrow CV = \Delta CS = EV$

Suppose $\partial x_1(p, w)/\partial w = 0$ on the relevant range. Use $h_1(p, u) = x_1(p, e(p, u))$ and differentiate in u :

$$\frac{\partial h_1(p, u)}{\partial u} = \frac{\partial x_1}{\partial w} \cdot \frac{\partial e(p, u)}{\partial u} = 0.$$

- So h_1 does not depend on u at all: $h_1(p, u^0) = h_1(p, u^1) = x_1(p, w)$ for every p on the path.
- The three areas are integrals of the *same* function:

$$-CV = \int_{p_1^0}^{p_1^1} h_1(t, u^0) dt = \int_{p_1^0}^{p_1^1} x_1(t, w) dt = \int_{p_1^0}^{p_1^1} h_1(t, u^1) dt = -EV. \blacksquare$$